Assignment - 4

Q1 Consider building a CSMA/CD network running at 10 Mbps over a cable with no repeats. If the signal speed in the cable is 10^6 km/sec & minimum frame size is 1500 bytes then what is the cable length.

Q2 A group of N stations share 50 Kbps slotted Aloha channel. Each station outputs a 500 bits frame on an average of once 5000 msec, even if previous one has not been sent. What is the maximum value of N?

Q3 A n/w with CSMA/CD protocol in the MAC layer is running a 1 Gbps over a 1 km cable with no repeaters. The signal speed in the cable is 2×10^8 m/sec. The min frame size for this n/w should be -

Q4 measurement of a slotted Aloha channel with an infinite number of users shown that 10% of the slots are idle.

(a) What is channel load, G?
(b) What is the throughput
(c) Is the channel underloaded or overloaded

Q5 A pure Aloha n/w transmits 200 bit frames on a shared channel of 200 Kbps. What is the requirement to make this frame collision free?

Scanned by CamScanner

Q6 A pure Aloha n/w transmits 200-bit frames on a shared channel of 200 kbps. What is the throughput if the system (all stations together) produces (i) 1000 frames/sec?
(ii) 500 frames/sec?
(iii) 250 frames/sec?

Q7 A slotted Aloha n/w transmits 200-bit frame using a shared channel with a 200 kbps bandwidth. Find the throughput if the system (all stations together) produces 500 frames/sec?

Q8 A n/w using CSMA/CD has a bandwidth of 10 mbps. If the maximum propogation delay is 25.6 μsec. What is the min size of the frame?

Scanned by CamScanner